

Program : Diploma in Information Technology	
Course Code : 5263	Course Title: Server Administration
Semester : 5	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- To provide the basic concepts of server administration, networking, FTP and print servers.
- To provide concepts for configuring samba and NFS file servers.

Course Pre-requisites:

Topic	Course Code	Course Title	Semester
Basics of networking		Fundamentals of network administration	3
Basic knowledge of System administration		System administration	4

Course Outcomes:

On completion of the course, students will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Outline basic concept of server administration and SSH software packages	15	Understanding
CO2	Illustrate basic concept of administrating network, IP addressing and network interfaces.	14	Understanding
CO3	Summarize basic concepts of print server, web server and FTP server	15	Understanding
CO4	Summarize basic concepts of samba and NFS file server	14	Understanding
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

On completion of the course, students will be able to:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Outline basic concept of server administration and SSH software packages.		
M1.01	Outline system administration skills	1	Understanding
M1.02	Classify the server package categories	1	Understanding
M1.03	Explain server configuration	1	Understanding
M1.04	Classify Linux security techniques	1	Understanding
M1.05	Explain server monitoring	1	Understanding
M1.06	Classify tools to monitor the server	1	Understanding
M1.07	Outline SSH	1	Understanding
M1.08	Explain SSH tools in client server software packages	1	Understanding
M1.09	Illustrate OpenSSH server services	1	Understanding
M1.10	Illustrate SSH client tools	2	Understanding
M1.11	Explain SSH for remote login and remote execution	2	Understanding
M1.12	Relate passwordless and key based authentication	2	Understanding
Contents: System administration skills-Server package categories-Configure the server-Starting the server administration-Linux security techniques-Monitor the server-Tools to monitor the server – SSH-SSH tools in client server software package-Open SSH server services-Using SSH client tools-SSH for remote login and execution – Passwordless-Key based authentication.			

CO2	Illustrate basic concept of administrating network, IP addressing and network interfaces.		
M2.01	Compare wired and wireless network	1	Understanding
M2.02	Explain Linux system to configuring network	1	Understanding
M2.03	Explain network configuration in Linux	1	Understanding
M2.04	Summarize checking network interfaces	1	Understanding
M2.05	Experiment with how to configure network interfaces	1	Understanding
M2.06	Explain how to set IP address manually	1	Understanding
M2.07	Explain IP address aliases	1	Understanding
M2.08	Illustrate setting routes	1	Understanding
M2.09	Outline network address, netmask and gateway	2	Understanding
M2.10	Explain configuring network proxy connection	2	Understanding
M2.11	Explain configuring Linux as a DHCP server, DNS, Router	2	Understanding
	Series Test – I	1	

Contents: Wired and wireless network-Linux system to configure network-Network configuration in Linux-Checking network interfaces-configure network interfaces-setting IP address manually, IP address aliases-setting routes-network address – netmask – gateway-configuring networking proxy connection-configuring Linux as a DHCP server-configuring Linux as a DNS- configuring Linux as arouter.

CO3	Summarize basic concepts of print server, web server and FTP server		
M3.01	Classify the common Unix printing system	1	Understanding
M3.02	Explain CUPS printing	1	Understanding
M3.03	Explain how to add a printer automatically	1	Understanding
M3.04	Summarize printer using CUPS administration	1	Understanding
M3.05	Summarize local printer, remote printer and SMB printer	3	Understanding
M3.06	Outline host and queue	1	Understanding
M3.07	Outline LAMP server and Apache web server	1	Understanding
M3.08	Explain how to Install Apache	1	Understanding
M3.09	Summarize Starting apache and Securing apache	2	Understanding
M3.10	Outline FTP and FTP server	1	Understanding
M3.11	Summarize setting up user access	1	Understanding

M3.12	Illustrate Allowing uploading	1	Understanding
Contents: Print server: common Unix printing system-CUPS printing, adding a printer automatically-configure a printer using CUPS administration-configure local printer-and configure remote printer-host and queue-SMB printer. Web server: Apache web server-Installing Apache-Starting apache-securing apache. FTP server: FTP- Configuring FTP server-setting up user access-allowing uploading			
CO4	Summarize basic concepts of samba and NFS file server		
M4.01	Outline samba	1	Understanding
M4.02	Explain samba Installation	1	Understanding
M4.03	Classify type of samba packages	1	Understanding
M4.04	Compare smb and nmb	1	Understanding
M4.05	Explain how to starting samba services	1	Understanding
M4.06	Classify the security features in samba	1	Understanding
M4.07	Outline port using samba	1	Understanding
M4.08	Explain how to configure samba	1	Understanding
M4.09	Explain accessing samba shares in Linux	2	Understanding
M4.10	Outline NFS server	1	Understanding
M4.11	Construct NFS server installation	1	Understanding
M4.12	Summarize starting NFS file system	1	Understanding
M4.13	Summarize sharing NFS file system	1	Understanding
	Series Test – II	1	
Contents: Samba: samba-Installing Samba-Types of samba package-smb and nmb-Starting samba services-Security features in samba-Port using samba-Configuring samba-smb.conf file-accessing samba shares in Linux NFS file server: NFS server-Installing an NFS server-starting the NFS file system-Sharing NFS file system			

Text / Reference

T/R	Book Title / Author
T1	Christopher Negus “Linux Bible” the comprehensive tutorial resources, Ninth Edition

R1	Richard Petersen “Linux: The Complete Reference”, Sixth Edition
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Online Resources

Sl.No	Website Link
1	https://www.tuxcademy.org/download/en/adm1/adm1-en-manual.pdf
2	https://www.tecmint.com/20-linux-yum-yellowdog-updater-modified-commands-for-package-mangement/
3	https://access.redhat.com/documentation/en-us/red_hat_enterprise_linux/6/html/deployment_guide/s2-users-cl-tools